Take-Home Assignment: Build an Agentic AI Contributor for Open-Source Go Projects

Objective

Build an **agentic AI platform** that can work on issues from open-source Go projects and generate production-quality code changes.

The goal is to evaluate how you think about agents, tools, repository understanding, code modification, validation, and extensibility.

You may use any LLM or AI coding tool, but we expect to see the system/framework you built around it — not just a one-shot prompt or thin wrapper.

Approved Projects

Choose one repository from the list below:

1. `gin-gonic/gin`
   https://github.com/gin-gonic/gin

2. `spf13/cobra`
   https://github.com/spf13/cobra

3. `go-playground/validator`
   https://github.com/go-playground/validator

4. `golangci/golangci-lint`
   https://github.com/golangci/golangci-lint

Pick a small or medium issue. Avoid issues that require large architectural changes, security-sensitive changes, major rewrites, or unclear maintainer decisions.

Problem Statement

Build a system that can take a GitHub issue from one approved project and assist in solving it.

The system should inspect the repository, understand the issue, identify relevant files and context, plan a fix, modify the code, run relevant tests or checks, and generate a pull request title and body.

Actually opening a pull request is optional. A local branch, patch, or diff with a PR summary is sufficient.

The exact architecture is up to you. You may use simple rules, repository maps, code search, embeddings, RAG, Claude/Cursor-style project rules, tool-based agents, or any other approach.

How We Will Validate

We may validate your system by running it against selected GitHub issues from the approved repositories, including issues that already have corresponding pull requests.

We will compare the agent's generated changes with the actual accepted PRs to understand how well the system:

- Identifies the right files
- Produces relevant code changes
- Follows project conventions
- Runs appropriate validation
- Generates a reasonable PR summary

Deliverables

Submit a GitHub repository containing:

- Your agentic AI system
- A README with setup and run instructions
- Any required artifacts, configs, prompts, rules, indexes, or sample outputs needed for us to run the system on our machine

The submission should be easy to run and review.

Important Note

We are not expecting a production-grade autonomous coding agent.

A simple, thoughtful framework that can solve focused issues reliably is better than a complex system that is hard to understand or unreliable.